Huzaifa
23172030
Sec 'A'

Date

Outlook	Temperature	~~Hot~~ Humidity	Windy	Rain
OL	T	H	W	R
Sunny	Hot	high	Weak	No
Sunny	mild	high	strong	No
cloudy	Hot	high	Weak	Yes
Rainny	mild	high	weak	Yes
Rainny	Cool	Normal	weak	Yes
Rainny	Cool	Normal	strong	No
cloudy	Mild	Normal	strong	Yes
Sunny	cool	Normal	weak	No

STEP 1

From Data Set

Total Records = 8

Yes = 4

No = 4

$H(S) = -\frac{4}{8} \log_2(\frac{4}{8}) - \frac{4}{8} \log_2(\frac{4}{8})$

$H(S) = 1.000$

Info Gain

Outlook	Total	Yes	No	Entropy
Sunny	3	0	3	0.000
cloudy	2	2	0	0.000
Rainy	3	2	1	0.918

$W.E = 0 + 0 + \frac{3}{8}(0.918) = 0.344$

Signature

No.

Date

W.E = 0.919 0.344

IG(Outlook) = 1.000 - ~~0.918~~

IG(outlook) = ~~0.082~~ ~~0.08~~2 0.65

Temprature

Temprature	Total	Yes	No	Entrop
HoT	2	1	1	1.000
Mild	3	2	1	0.918
Cool	3	1	2	0.918

W.E = ~~2.83~~6 $\frac{3}{8}(1.)+\frac{3}{8}(0.918)+$

~~IG = $\frac{3}{8}$(1.000) 2.83~~6 $\frac{3}{8}$(0.9

IG = ~~1.83~~6 1.000-938

IG(T) = 0.615

Humidity

Humidity	Total	Yes	No	Entrop
High	4	2	2	1.000
Normal	4	2	2	1.000

W.E = ~~2.000~~ $\frac{4}{8}(1) + \frac{4}{8}(1) =$

IG (Humidity) = 1.000 - 1.000

IG (Humidity) = 0.000

Signature

Windy

Windy	Total	Yes	No	Entoopy
Weak	5	3	2	0.971
Strong	3	1	2	0.918

W.E = ~~1.889~~ $\frac{5}{8}(0.971)+\frac{3}{8}(0.918)$

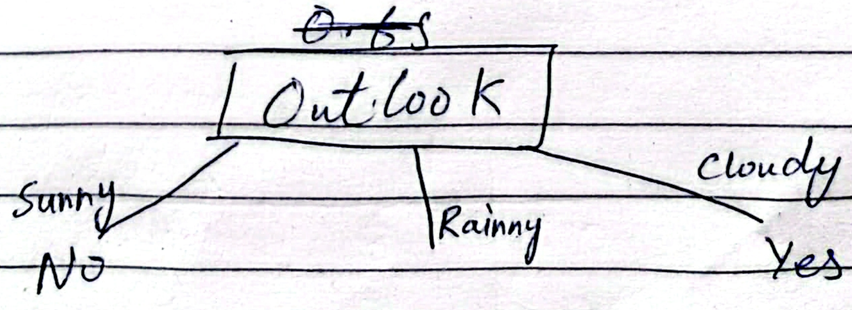
IG(Windy) = 1.000 − 0.951

IG(Windy) = 0.049

Compaøision	Entoopy
outlook	0.655
Temprature	0.615
Humidity	0.00
Windy	0.049

Outlook has the highest Entoopy.

So, outlook selected as Root Node



CamScanner